AI-Guided Cloud Migration: Any Cloud to Huawei Cloud



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What AI Can and Cannot Do for Your Migration

Sharing of capabilities, not a formal delivery commitment

AI guidance for cloud-to-cloud migration | Huawei Cloud

Central Note

AI accelerates cloud migration by automating discovery, mapping, and planning. It does not replace your judgment, your team's expertise, or customer approval. Use it as a powerful assistant — not as an autopilot.

1. Your Data Stays Yours

Security / Compliance / Protection

- AI does not need your real customer data
- Use simulated, test, or anonymized data
- Never upload credentials, keys, or sensitive data

2. AI Assists, You Decide

Reference / Human Review

- AI plans, analyzes, and writes scripts
- Your team reviews and approves everything
- Test in staging before production migration

3. Get Customer Approval

Authorization / Approval

- The customer decides if AI is used in their project
- Without approval, do not enter customer details into AI
- Respect security, compliance, and confidentiality

4. When to Be Extra Careful

Prudent Evaluation / Manual Process

- Highly sensitive systems need extra review
- Financial core, defense, or classified workloads
- Consider manual migration for strict sovereignty rules

AI-assisted migration improves efficiency; it does not replace human review or customer authorization. Always validate in your specific environment before production cutover.

Why Moving Between Clouds Is Hard



What Makes It Difficult

Every cloud provider has its own way of naming and configuring services. What AWS calls an EC2 instance, Huawei Cloud calls an ECS. Security rules, networking, and storage all work differently. Mapping everything by hand takes weeks and is easy to get wrong.



How AI Changes This

AI can read your source cloud, understand what you have, and automatically map each resource to its Huawei Cloud equivalent. It turns weeks of manual analysis into a conversation. You describe what you want — AI handles the technical details.



What You Will See

A real example: a WordPress workload on AWS (servers, database, storage) migrated to Huawei Cloud. You will see how AI discovers, plans, and executes the migration — and what your role is at each step.

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How Your AI Assistant Approaches the Migration

Step 1: Discover What You Have

Give AI read-only access to your source cloud. It will list every server, database, storage bucket, network, and security rule — and map the dependencies between them. No manual inventory needed.

Example prompt: "Scan my AWS account and list all resources with their dependencies."

Step 2: Map to Huawei Cloud

AI matches each source resource to its Huawei Cloud equivalent. It considers functionality, performance, and cost. You review the mapping and adjust anything that does not fit.

Example prompt: "Map these AWS resources to Huawei Cloud equivalents and explain each choice."

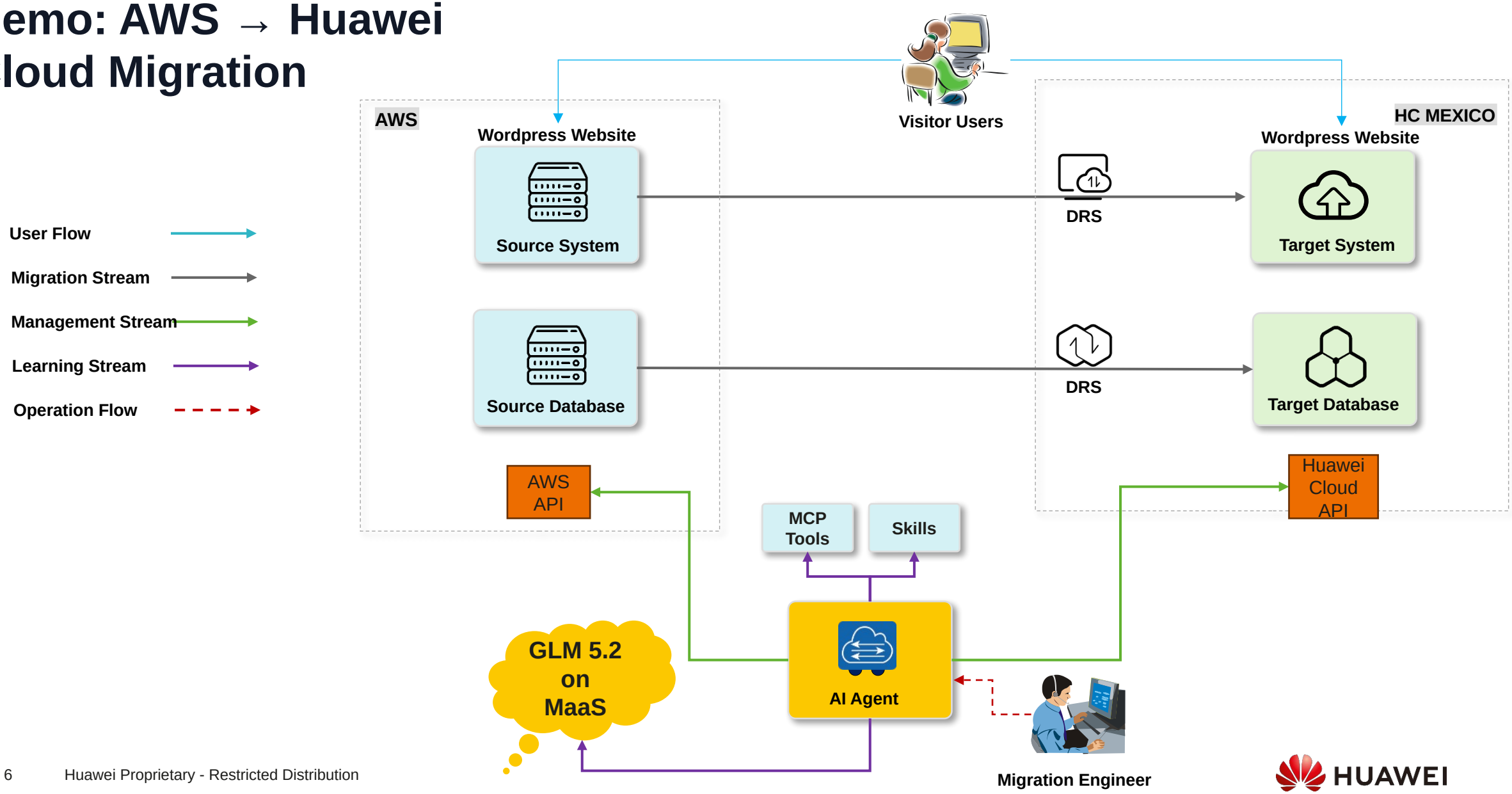
Step 3: Plan the Migration

AI creates a step-by-step migration plan based on dependencies. It identifies what must move first, estimates timing, and flags risks. You approve the plan before anything moves.

Example prompt: "Create a migration plan with phases, timing, and risk notes."

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Demo: AWS → Huawei Cloud Migration



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Service Mapping: Source Cloud to Huawei Cloud

Source Cloud Service	Huawei Cloud	Migration Tool	What Happens
Virtual Servers (EC2)	ECS	SMS	Server migrated with disks and network
Managed Database (RDS)	RDS	DRS	Database replicated with minimal downtime
Object Storage (S3)	OBS	OMS	Files copied with integrity verification
Virtual Network (VPC)	VPC	Terraform	Network recreated with same topology
Subnets	Subnets	Terraform	Public and private subnets recreated
Security Groups	Security Groups	Terraform	Firewall rules mapped automatically
Internet Gateway	VPC Gateway	Terraform	Internet access configured
Route Tables	Route Tables	Terraform	Routing rules recreated
Key Pairs	KPS	Terraform	SSH keys managed
Disk Volumes (EBS)	EVS	SMS	Disks migrated with the server
Monitoring	CES	Manual	Set up after migration

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Your Migration Toolkit

Server Migration Service

What it does:
- Migrates servers from any source
- Supports Linux and Windows
- Automatically configures network
- Handles disk conversion

Ask AI about it:
"Which servers should I migrate with SMS?"

Data Replication Service

What it does:
- Migrates databases with minimal downtime
- Full and incremental sync
- Supports MySQL, PostgreSQL, and more
- Online migration — no long outage

Ask AI about it:
"How do I set up DRS for my database?"

Object Storage Migration

What it does:
- Moves files between clouds
- Verifies every file arrived intact
- Handles large-scale transfers
- Works with 11+ source clouds

Ask AI about it:
"How long will OMS take for my data?"

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AI Writes Your Infrastructure Code

What AI Reads from Your Source Cloud

AI scans your source cloud and identifies:
- Servers and their configurations
- Databases and their versions
- Storage buckets and their contents
- Networks, subnets, and security rules
- All dependencies between resources

Example prompt:
"Read my AWS Terraform and generate the Huawei Cloud equivalent."



What AI Produces for Huawei Cloud

AI generates ready-to-use Terraform code:
- Network module (VPC, subnets, security)
- Server migration module (SMS templates)
- Database migration module (DRS jobs)
- Database migration module (DRS jobs)
- Storage migration module (OMS tasks)

You review the code, adjust as needed, and run it. AI explains every line.



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How AI, Your Team, and Huawei Cloud Work Together

What AI Does

- Analyzes your source cloud
- Generates Terraform code
- Executes migration via SMS, DRS, OMS
- Monitors progress and handles errors

What Your Team Does

- Reviews and approves AI's plans
- Adjusts configurations as needed
- Makes go/no-go decisions
- Controls the migration flow

What Gets Better Each Time

- AI learns from each migration
- Service mapping becomes more accurate
- Common patterns are remembered
- Future migrations go faster

The Workflow: You request → AI analyzes source → AI proposes plan → You review → AI executes → You verify

AI runs on Huawei Cloud MaaS with MCP tools that connect to both your source cloud and Huawei Cloud APIs.
Your team stays in control at every step. AI handles the repetitive, technical work.

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What You Gain with AI-Guided Migration



Save Time and Effort

AI cuts analysis and planning from weeks to hours. It eliminates manual errors in service mapping. Your team focuses on decisions, not spreadsheets.



Reduce Migration Costs

AI recommends the most cost-effective Huawei Cloud resources for your workloads. It identifies what can be optimized during migration.



Catch Problems Early

AI spots risks, incompatibilities, and hidden dependencies before you migrate. No surprises during cutover.

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Thank you.

Bring digital to every person, home and organization for a fully connected, intelligent world.

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